

B Tech Lab Experiments

Experiment 6

AIM OF THE EXPERIMENT: To determine the strength of the KMnO_4 solution using standard oxalic acid solution

APPARATUS USED: Beaker, test tube, conical flask, electronic balance, Bunsen burner, funnel, pipette, burette, wire gauze, burette stand, spatula, tripod stand.

CHEMICALS USED: Oxalic acid, dil. H_2SO_4 , Potassium permanganate (KMnO_4) solution

THEORY: Titration is a common laboratory method of qualitative chemical analysis that can be used to determine the unknown concentration of a solution (**analyte**). The basis of this process is the reaction between the analyte and a solution of known concentration (**standard solution**).

Primary standard: A highly purified compound that serves as the reference material for titration or another type of quantitative analysis. **NOTE:** Concentration of a primary standard solution does not change with time. Example- Oxalic acid ($\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$), Potassium dichromate ($\text{K}_2\text{Cr}_2\text{O}_7$) etc.

Secondary standard: A compound whose purity has been determined by chemical analysis (standardization) and concentration changes with time. Example: Potassium permanganate (KMnO_4), Sodium thiosulfate ($\text{Na}_2\text{S}_2\text{O}_3$) etc.

Concentration of a solution is defined as the amount of a solute present in a definite volume of the solvent. Concentration of a solution can be expressed in different ways.

- Normality: Normality of a solution is defined as the number of gram equivalent of solute per litre of the solution. It is denoted by 'N'.

$$\text{Normality} = \frac{\text{Number of gram equivalent(s) of solute}}{\text{Volume of solution (in litre)}}$$

- Molarity: Molarity of a solution is defined as the number of moles of the solute per litre of the solution. It is denoted by 'M'.

$$\text{Molarity} = \frac{\text{Number of moles of solute}}{\text{Volume of solution (in litre)}}$$

End point of a titration:

The endpoint of a titration is the point at which the reaction between the titrant and the analyte becomes complete. Generally, the endpoint of a titration is determined using indicators. In some cases, either the reactant or the product can serve as the indicator.

Titration can be classified as:

- Acid-Base Titrations or Acidimetry and Alkalimetry
- Oxidation-Reduction Titrations or Redox Titrations
- Precipitation Titrations
- Complexometric Titrations

Here we will learn about **Redox titrations**.

Oxidation-Reduction Titrations or Redox Titrations

- The titration based on oxidation and reduction reaction between the titrant and analyte is called **Redox titration**. **Oxidation** is the process of the addition of oxygen or removal of hydrogen/electron and **reduction** involves the process of addition of hydrogen/electrons or removal of oxygen. **Oxidizing agents** are substances that gain one or more electrons and are reduced. **Reducing agents** are substances that lose one or more electrons and are oxidized. That is, oxidizing agents are electron acceptors, and reducing agents are electron donors.
- In redox systems, the titration method can be adopted to determine the strength of a reductant/oxidant using a redox sensitive indicator. Redox titrations involving potassium permanganate are called permanganometric titrations. In these reactions, MnO_4^- ions act as the **self-indicator**.

Preparation of standard solution of Oxalic acid [250 ml M/10 (0.1 M) solution]

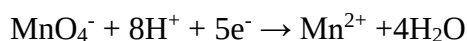
The molecular mass of crystalline oxalic acid is, $\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O} = 126$

Weight of oxalic acid crystals required to prepare 1000 ml of 1 M solution = 126 g

Therefore, weight of oxalic acid required to prepare 250 ml 0.1 M solution = $\frac{126}{1000} \times 250 \times 0.1 = 3.15g$

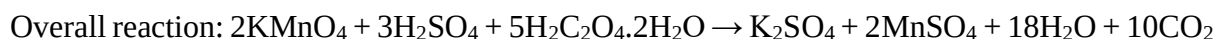
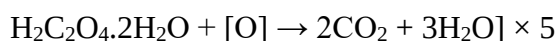
Determination of strength of KMnO_4 using standard solution of oxalic acid

In this titration KMnO_4 is the oxidizing agent and oxalic acid is the reducing agent. The reaction between potassium permanganate and oxalic acid is carried out in an acidic medium because permanganate ion in the acidic medium is a very strong oxidizing agent. Acidity is introduced by adding dil. H_2SO_4 .

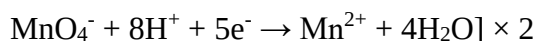


No other indicators are used to determine the endpoint, because KMnO_4 acts as the self-indicator. Permanganate (MnO_4^-) ion has a dark purple colour. In an acidic medium, MnO_4^- is reduced to colourless manganous (Mn^{2+}) ions. On reaching the end point, the addition of the last single drop of permanganate imparts a light pink colour to the solution. The chemical reaction that takes place during titration can be represented by the chemical equation.

Molecular equation



Ionic equation



PROCEDURE:

For preparation of standard solution of Oxalic acid [250 ml M/10 (0.1 M) solution]

- Using an electronic balance, first weigh exactly 3.15g of oxalic acid crystals in a weighing bottle.
- Transfer these into a 250ml beaker.

- Dissolve the oxalic acid crystals in the beaker containing distilled water (about 100 ml) by gentle stirring with a clean glass rod.
- When the oxalic acid crystals in the beaker are completely dissolved, transfer the entire solution from the beaker into a 250ml volumetric flask through a funnel and a glass rod.
- Add enough distilled water to the volumetric flask so that the level is just below the calibration mark on it.
- Stopper the measuring flask and shake gently to make the solution uniform throughout.

Determination of strength of given KMnO_4

- Rinse and fill the burette with the given KMnO_4 solution and set the initial burette reading as zero.
- Clamp it vertically to the burette stand.
- Rinse the pipette with the given oxalic acid solution.
- Then pipette out 25 ml of the oxalic acid solution into a conical flask and add 2 ml of conc. H_2SO_4 followed by 50 ml water into it.
- Heat the contents of the conical flask to $60-70^\circ\text{C}$.
- Titrate this hot solution against the KMnO_4 solution taken in the burette till a faint pink colour persists for about 30s, which is the end point of the titration.
- Note down the final burette reading.
- Repeat the titration until concordant values are obtained.

OBSERVATION:

No. of obs.	Volume of oxalic acid taken (ml)	Burette reading (ml)		Volume of KMnO_4 required (ml)	Concordant volume (ml)
		Initial	Final		
1.	25				B
2.	25				
3.	25				

Calculation:

V_1 = Vol. of oxalic acid solution taken = 25ml

S_1 = Strength of oxalic acid solution = 0.1 M

V_2 = Concordant volume of KMnO_4 solution = **B**

S_2 = Strength of KMnO_4 solution =?

From the balanced chemical equation, it is clear that 2 moles of KMnO_4 reacts with 5 moles of oxalic acid.

According to the molarity equation,

$$\frac{\text{Molarity of KMnO}_4 \times \text{Volume of KMnO}_4}{\text{Molarity of oxalic acid} \times \text{Volume of oxalic acid}} = \frac{\text{No. of moles of KMnO}_4}{\text{No. of moles of oxalic acid}} = \frac{2}{5}$$

$$S_2 = \frac{2S_1V_1}{5V_2}$$

Conclusion:

Strength of KMnO_4 solution = _____ M

Strength of KMnO_4 in g/L = Strength of $\text{KMnO}_4 \times$ molar mass of KMnO_4

$$= S_2 \times 158.034$$

NOTE:

Calculation can be done in terms of normality too using the formula $V_1S_1=V_2S_2$, as 1 equivalent of oxalic acid reacts with 1 equivalent of KMnO_4 .

Experiment 7

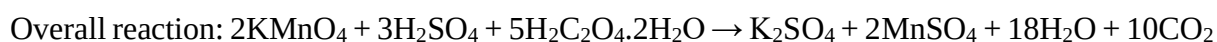
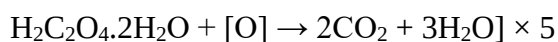
AIM OF THE EXPERIMENT: To determine the amount of iron (II) present in the supplied solution using standard KMnO_4 solution

APPARATUS AND CHEMICALS REQUIRED: Burette, pipette (20 mL), 250 mL conical flask, 250 mL volumetric flask, oxalic acid, potassium permanganate, sulfuric acid and phosphoric acid mixture

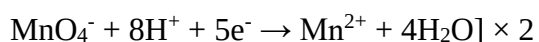
THEORY:

The titration of potassium permanganate (KMnO_4) against supplied iron solution (say Mohr's salt, $\text{FeSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$) is an example of redox titration (oxidation-reduction titrations). Potassium permanganate is a strong oxidant in the presence of sulfuric acid. Here supplied iron solution is titrated against potassium permanganate in the presence of acidic medium (sulfuric acid). Acidic medium is necessary in order to prevent precipitation of manganese oxide. Here KMnO_4 acts as a self-indicator and this titration is called permanganate titration.

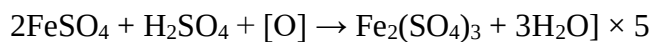
This process was presented by Mabride in 1912. When oxalic acid acidified with dil. H_2SO_4 is titrated with potassium permanganate solution at 60-70 °C temperature, an oxidation-reduction reaction takes place between these substances. The chemical reaction is-



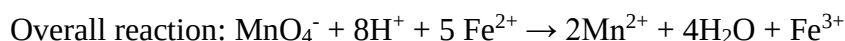
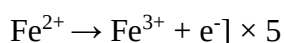
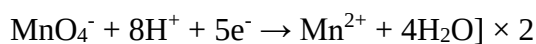
Ionic equation



When solution of Fe^{2+} ion acidified with dil. H_2SO_4 is titrated with potassium permanganate solution at room temperature, an oxidation-reduction reaction takes place between these substances. The chemical reaction is-



Ionic equation



PROCEDURE:

A. Preparation of standard solution of Oxalic acid [250 ml N/10 (0.1 N) solution]

The molecular mass of crystalline oxalic acid is, $\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O} = 126$

Equivalent weight of oxalic acid = $126/2 = 63\text{g}$

1000 ml of 1 N Oxalic acid solution contains 63 g oxalic acid

$$250 \text{ ml } 0.1 \text{ N Oxalic acid solution contains} = \frac{63}{1000} \times 250 \times 0.1 = 1.575\text{g}$$

Weigh accurately 1.575 g of oxalic acid in a chemical balance (or electronic balance) and transfer it to a 250 mL volumetric flask. Add 100 mL distilled water and shake vigorously until all the crystal is dissolved. Add again distilled water upto the mark of flask and it is labelled as 0.1N oxalic acid solution.

B. Determination of strength of KMnO_4 by titrating it against standard Oxalic acid solution

- Rinse and fill the burette with the given KMnO_4 solution and set the initial burette reading as zero.
- Clamp it vertically to the burette stand.

- Rinse the pipette with the given oxalic acid solution.
- Then pipette out 25 ml of the oxalic acid solution into a conical flask and add 2 ml of conc.H₂SO₄ followed by 50 ml water into it.
- Heat the contents of the conical flask to 60-70°C.
- Titrate this hot solution against the KMnO₄ solution taken in the burette till a faint pink colour persists for about 30s, which is the end point of the titration.
- Note down the final burette reading.
- Repeat the titration until concordant values are obtained.

FOLLOW EXPERIMENT 6

C. Estimation of Fe²⁺ present in the supplied iron solution

- Pipette out 25 mL iron (II) solution in a 250 mL conical flask.
- Add 20 mL (one test tube) H₂SO₄ and H₃PO₄ acid mixture.
- Titrate the resulting solution with standard KMnO₄ solution until a faint pink colour appears which is the end point of the titration.

OBSERVATION:

Table 1. Titration of oxalic acid against KMnO₄ solution					
No. of obs.	Volume of oxalic acid taken (mL)	Burette reading (ml)		Volume of KMnO₄ required (ml)	Concordant volume (ml)
		Initial	Final		
1.	25				B
2.	25				
3.	25				

Table 2. Titration of supplied iron solution with KMnO ₄					
No. of obs.	Vol. of supplied iron solution (mL)	Burette reading (ml)		Volume of KMnO₄ required (ml)	Concordant volume (ml)
		Initial	Final		
1.	25				X
2.	25				
3.	25				

CALCULATION:

A. Strength of KMnO₄ solution:

We know,

$$V_1 S_1 = V_2 S_2$$

$$S_2 = \frac{V_1 S_1}{V_2}$$

$$= \text{_____ N}$$

Here, V₁= volume of oxalic acid

S₁ = strength of oxalic acid

V₂ = volume of KMnO₄ required = final burette reading

S₂ = strength of KMnO₄

B. Determination of the amount of iron (II) present in the supplied solution

1000 mL 1N KMnO₄ solution $\equiv$ Atomic wt. of Fe = 55.85 g mol⁻¹

1mL 0.1 N KMnO₄ solution $\equiv \frac{55.85}{1000} \times 0.1 \text{ g Fe}^{2+} = 0.005585 \text{ g Fe}$

X mL 0.1 N KMnO₄ solution $\equiv 0.005585 \times X \text{ g Fe}$

Amount of Fe²⁺ in g L⁻¹ $= \frac{0.005585 \times X \times 1000}{25} = \text{_____ g}$

CONCLUSION:

The amount of Fe²⁺ estimated in the supplied iron solution is g/L.

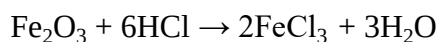
Experiment 8

AIM OF THE EXPERIMENT: To determine the amount of Iron (III) present in the supplied solution using standard $K_2Cr_2O_7$ (potassium dichromate).

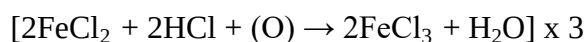
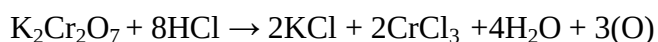
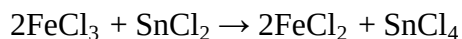
APPARATUS AND CHEMICALS USED: Burette, pipette (20 mL), 250 mL conical flask, 250 mL volumetric flask, potassium dichromate solution ($K_2Cr_2O_7$), sulfuric acid (H_2SO_4) and phosphoric acid (H_3PO_4) mixture, conc. HCl, $SnCl_2$, $HgCl_2$, barium diphenylamine sulfonate indicator.

THEORY:

Hematite is an oxide ore of iron, existing as ferric oxide (Fe_2O_3). On treating with HCl, ferric oxide dissolves in the acid as ferric chloride.

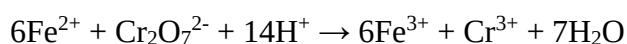
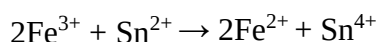


Since iron in $FeCl_3$ is already in the highest possible stable oxidation state (Fe^{3+}), it cannot be oxidized further by potassium dichromate solution, an oxidizing agent. Therefore, it is first reduced to ferrous state (Fe^{2+}) by stannous chloride in presence of hydrochloric acid ($SnCl_2 + HCl$) and then titrated against potassium dichromate ($K_2Cr_2O_7$).



Overall reaction: $K_2Cr_2O_7 + 14HCl + 6FeCl_2 \rightarrow 2KCl + 2CrCl_3 + 6FeCl_3 + 7H_2O$

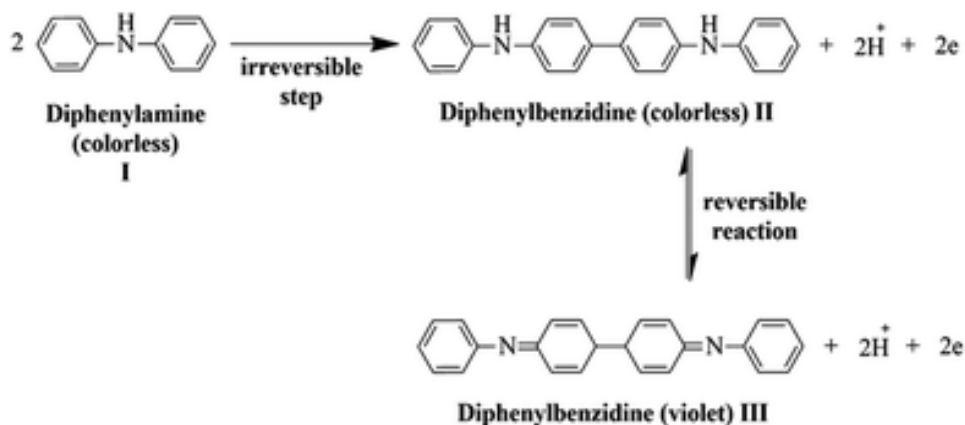
Ionic equation:



A slight excess of stannous chloride is added to ensure complete reduction of $FeCl_3$ to $FeCl_2$. The excess of $SnCl_2$ added is destroyed by adding strong solution of mercuric chloride, $SnCl_2$ is a good reducing agent and can react with potassium dichromate (oxidizing agent) if excess $SnCl_2$ is not neutralized.

Barium diphenylamine is used as an indicator to detect the end point of the

titration. The diphenylamine (I) undergoes oxidation first into a colourless diphenylbenzidine (II) which is the real indicator and is reversibly further oxidized to diphenylbenzidine violet (III) at end point.



NOTE:

- Role of phosphoric acid

Formation of a colorless complex with the produced ferric ions leading to:

1. Decreasing the molar concentration of Ferric and hence reduces the actual potential of (Fe³⁺ /Fe²⁺) system so that Fe²⁺ ion will be oxidized before the indicator.

(The oxidation potential of Fe³⁺ /Fe²⁺ = 0.77 volt)

2. Removing the dark yellow color of Fe³⁺ ion giving a clearer color change.

- Diphenylamine is only able to function as indicator in this reaction when phosphoric acid is present in the solution.

PROCEDURE:

Preparation of 0.1N potassium dichromate solution

The molecular mass of potassium dichromate K₂Cr₂O₇ is = 294.185

Equivalent weight of K₂Cr₂O₇ = 294.185/6 = 49.03g

1000 ml of 1 N K₂Cr₂O₇ solution contains 49.03 g K₂Cr₂O₇

250 ml 0.1 N K₂Cr₂O₇ solution contains = $\frac{49.03}{1000} \times 250 \times 0.1 = 1.225\text{g}$

Weigh accurately 1.225 g of potassium dichromate in a chemical balance (or

electronic balance) and transfer it to a 250 mL volumetric flask. Add 100 mL distilled water and shake vigorously until all the crystal is dissolved. Add again distilled water upto the mark of flask and it is labelled as 0.1 N $\text{K}_2\text{Cr}_2\text{O}_7$ solution.

Estimation of Fe^{3+} by standard 0.1N potassium dichromate solution

1. Pipette out 25mL of supplied Fe (III) solution in a conical flask.
2. Add 3 mL ($1/5^{\text{th}}$ test tube) conc. HCl and heat to boiling.
3. To this hot solution add SnCl_2 solution drop by drop with shaking till the solution is just colourless.
4. Cool the solution immediately under the tap water to room temperature and add 10 mL ($1/2$ test tube) HgCl_2 solution all at a time when a slight silky white ppt. appears. (If grey or black ppt. appears, discard the whole solution and start afresh).
5. Add about 100 mL (5 test tube full) water, 20 mL (1 test tube full) $\text{H}_3\text{PO}_4 + \text{H}_2\text{SO}_4$ mixture and 4-5 drops of barium diphenylamine sulfonate indicator.
6. Titrate the resulting solution with standard $\text{K}_2\text{Cr}_2\text{O}_7$ solution taken in a burette till the colour of the solution turns just violet which is the end point of the titration.
7. Note down the end point. Repeat the titration till the concordant volume is obtained.

OBSERVATION:

Table 2. Titration of supplied iron solution with KMnO_4					
No. of obs.	Vol. of supplied iron solution (mL)	Burette reading (ml)		Volume of $\text{K}_2\text{Cr}_2\text{O}_7$ used (ml)	Concordant volume (ml)
		Initial	Final		
1.	25				V
2.	25				
3.	25				

CALCULATION:

Given,

Strength of $\text{K}_2\text{Cr}_2\text{O}_7$ solution = 0.1 N

1 ml of 1N $\text{K}_2\text{Cr}_2\text{O}_7$ solution $\equiv$ 0.05585g of Fe

V ml of 0.1N $\text{K}_2\text{Cr}_2\text{O}_7$ solution $\equiv$ $0.05585 \times 0.1 \times V$ g of Fe

Amount of iron in g/L = $\frac{0.05585 \times 0.1 \times V \times 1000}{25}$ g

CONCLUSION:

Amount of iron present in 1 litre of supplied solution = _____ g.